Translations of Graphs

Exam code: 0607

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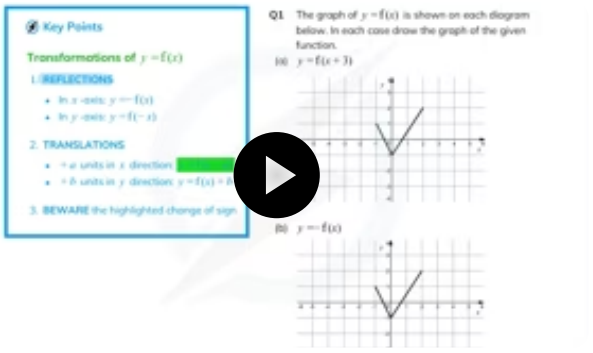

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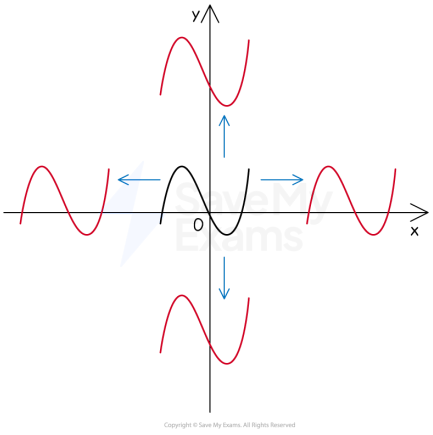
Help me understand

Translations of graphs

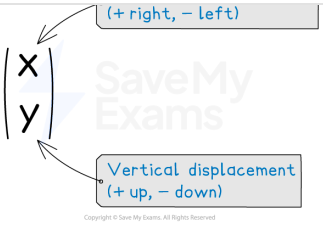
What are translations of graphs?

- The **equation** of a graph can be changed in certain ways
  - This has an effect on the **graph**
    - How a graph changes is called a **graph transformation**
- A **translation** is a type of graph transformation that **shifts (moves)** a graph (up or down, left or right) in the *xy* plane
  - The shape, size, and orientation of the graph remain unchanged



Examples of translations

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

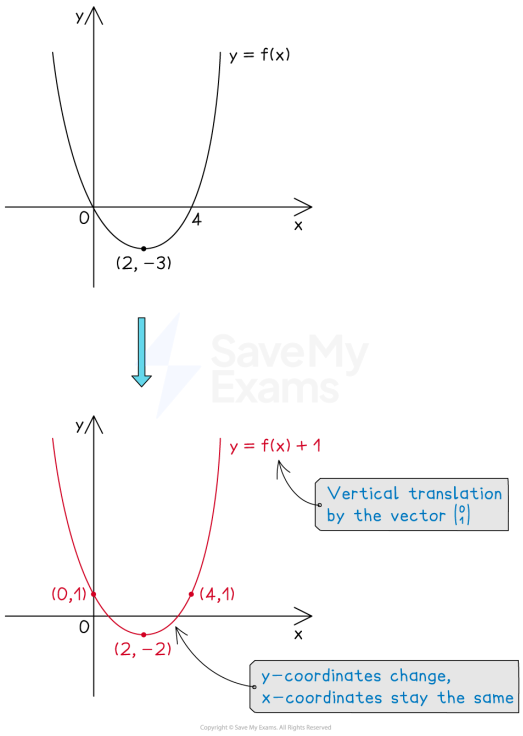


How do I translate graphs?

- Let $y = \mathrm{f}(x)$ be the **equation** of the **original graph**

Vertical translations: y=f(x) + a

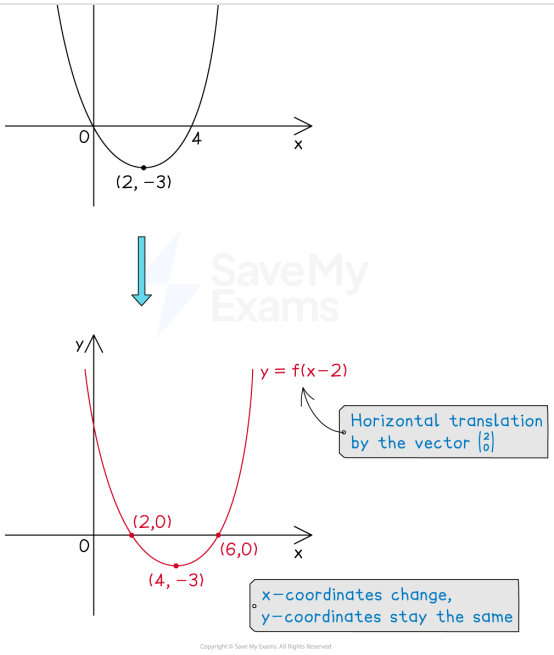
- $y = \mathrm{f}(x) + a$ is a **vertical translation** by the vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$
  - The graph moves **up for positive** values of a
  - The graph moves **down for negative** values of a
  - The ***x*-coordinates** stay the **same**



Example of a vertical translation

Horizontal translations: y=f(x + a)

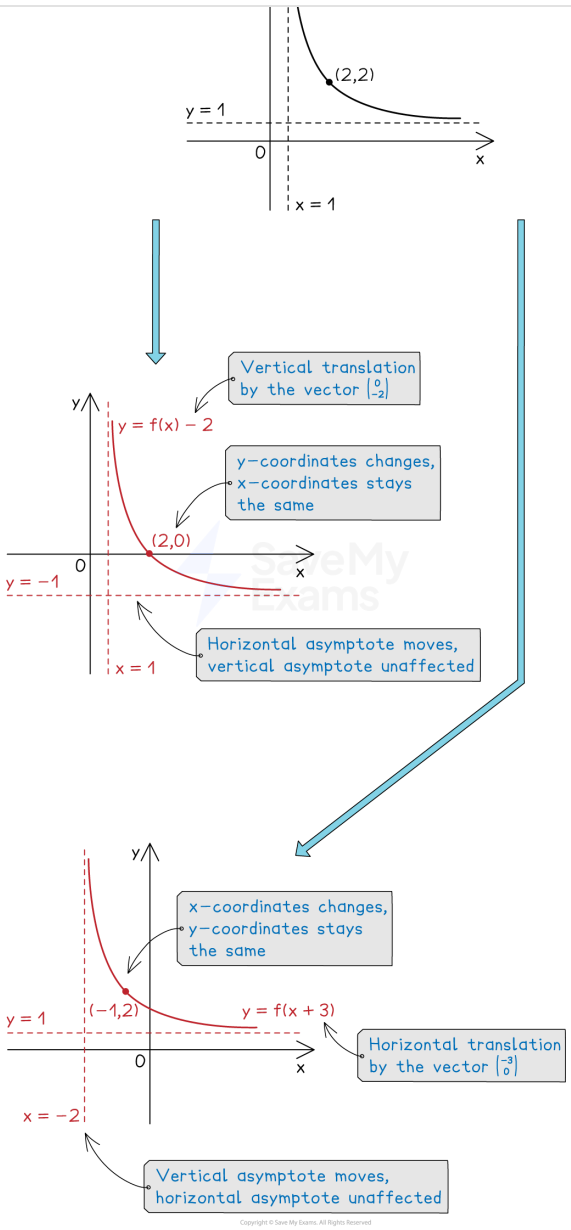
- $y = \mathrm{f}(x + a)$ is a **horizontal translation** by the vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$
  - The graph moves **left for positive** values of a
    - This is often the opposite direction to which people expect
  - The graph moves **right for negative** values of a
  - The ***y*-coordinates** stay the **same**



Example of a horizontal translation

What happens to asymptotes when a graph is translated?

- Any **asymptotes** of **f(x)** are also translated
  - An asymptote **parallel** to the **direction of translation** will **not** be affected



How does a translation affect the equation of the graph?

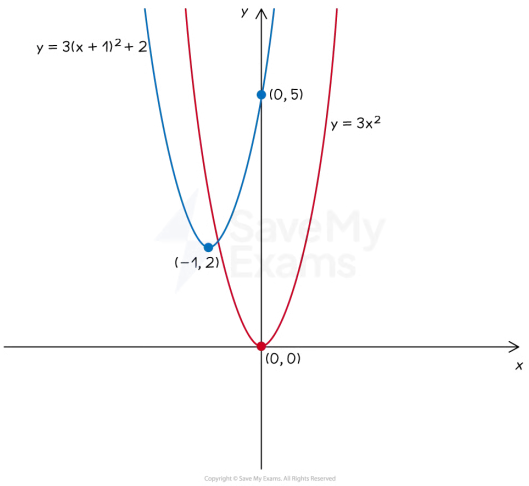
- For a **horizontal translation** $y = \mathrm{f}(x - a)$ of the graph $y = \mathrm{f}(x)$
  - $\boldsymbol{a}$ **is subtracted from** $\boldsymbol{x}$ throughout the equation
  - Every instance of x in the equation is replaced with $(x - a)$

- E.g. the graph $y = x^2 - 3x + 7$ undergoes a **translation** of **6 units** to the **right**
  - $y = \mathrm{f}(x)$ becomes $y = \mathrm{f}(x - 6)$
  - x is replaced throughout the equation by $(x - 6)$
    - $y = (x - 6)^2 - 3(x - 6) + 7$ is the **new** equation
  - The equation can be left in this form or expanded and simplified
    - $y = x^2 - 12x + 36 - 3x + 18 + 7$
    - $y = x^2 - 15x + 61$

- E.g. the graph $y = 4x^2 + 2x + 1$ undergoes a **translation** of **5 units down**
  - $y = f(x)$ becomes $y = f(x) - 5$
  - 5 is subtracted from the equation as a whole
    - $y = 4x^2 + 2x + 1 - 5$
  - The equation can be left in this form or simplified
    - $y = 4x^2 + 2x - 4$

How do I apply a combined translation?

- For a **horizontal translation** of p units and **vertical translation** of q units **combined**
  - $y = f(x)$ becomes $y = f(x - p) + q$
- E.g. the graph $y = 3x^2$ undergoes a **translation** of **2 units up** and **1 unit** to the **left**
  - $y = f(x)$ will become $y = f(x + 1) + 2$
  - x is replaced throughout the equation by $(x + 1)$
  - 2 is added to the equation as a whole
    - $y = 3(x + 1)^2 + 2$



- Note that when the equation is in the form $y = a(x - p)^2 + q$
  - the **vertex** is (p, q)
  - the value of a does not affect the vertex coordinates

Worked Example

The diagram below shows the graph of $y = f(x)$.

Sketch the graph of $y = f(x+3)$.

Answer:

The transformation of the graph is a horizontal translation with vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ (3 units to the left)

The x-coordinates of the points change (subtract 3 from each)
The y-coordinates of the points stay the same

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